

When a Safety Layer Stops Being One: Base-Point Staleness in Voltage-Constrained EV Charging

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Abstract—Projecting a charging set-point onto a linearised voltage constraint has become the standard way to make a learned or heuristic controller respect a distribution feeder’s limits. The linearisation has two inputs that are procured very differently: a Jacobian, computed from a network model, and a base point, which can only be measured. We separate them on two unrelated radial feeders, a medium-voltage IEEE 33-bus testbed and a 117-bus low-voltage village benchmark. A Jacobian wrong by a factor of two costs nothing. A base point that is never refreshed removes the protection altogether: violations return to within three percent of the unprojected rate in all twelve cells tested, while the layer reports no infeasibility whatsoever. The transition is abrupt, and it arrives at different ages on the two feeders. A second, opposite fault, under-estimated line impedance, buys a spotless violation record with curtailment. Safety layers need instrumenting, not outcome-checking.

Index Terms—Electric vehicle charging, power distribution, reinforcement learning, voltage control, voltage sensitivity.

I. INTRODUCTION

Electrifying road transport moves a large new load onto networks that were never planned around it. Strong urban distribution absorbs it comfortably. The weak radial feeders serving rural and semi-rural customers do not: terminal voltage there begins to sag at low penetration, and the classical remedies—reconductoring, a larger transformer, another lateral—are slow, disruptive and capital-intensive. Buying hosting capacity with control rather than with copper is therefore an appealing proposition, and it is the premise of most smart-charging and vehicle-to-grid work aimed at decarbonising transport without first rebuilding the network beneath it.

Control bought in place of copper needs an assurance attached, because the operator is now trusting software to keep voltage legal. The prevailing way of supplying one is a *safety layer*: a small optimisation, sitting between whatever chooses the set-points and the plant, that takes each request and returns the nearest request a linearised power-flow model certifies as admissible [1]–[6]. Its appeal is that it does not care what sits above it—a trained agent, a tariff heuristic, a rule—and that it converts a statistical claim about a policy into a per-step constraint on an action. In the field’s own taxonomy of safe reinforcement learning for power systems, projection of this kind is one of two techniques within one of the two top-level categories [7].

Underneath the layer sits a linearisation, and that linearisation has two inputs which the literature almost universally

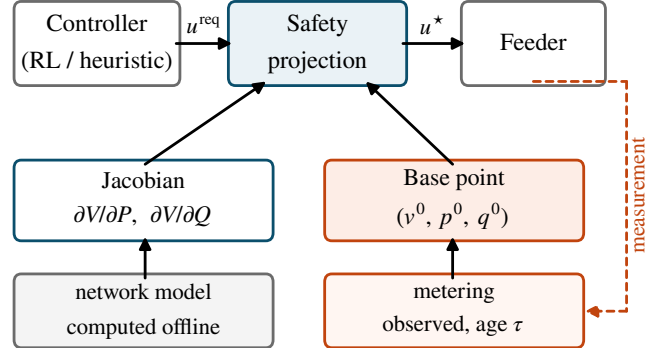


Fig. 1. The safety layer has two inputs, procured by different means. The Jacobian comes from a network model and can be recomputed offline at will; the base point can only be observed, at whatever cadence the metering provides. This paper perturbs each independently.

handles as a single object. The *Jacobian* $\partial V/\partial P$, $\partial V/\partial Q$ describes how the network responds; obtaining it requires an admittance model, and it changes only when the network does. The *base point*—the operating condition about which the Jacobian was taken—describes where the feeder currently is; obtaining it requires a measurement, and it changes continuously. One can be recomputed offline as often as one likes. The other cannot be computed at all (Fig. 1).

None of the work cited above reports how often its linearisation is refreshed. On a transmission system that omission would matter little. On distribution it matters a great deal: the state-estimation literature describes these networks as practically unobservable in real time [8], measurement-scarce, served by sensors with heterogeneous reporting rates, and easily tipped into unobservability by a communications fault. Distribution state estimation runs on cycles of minutes; advanced metering reports on a quarter-hour to an hour. A base point that is somewhat out of date is not a corner case on a distribution feeder. It is the ordinary condition.

A. What is, and is not, claimed here

We are explicit about what is not ours. That measurement can substitute for model accuracy is established in at least three literatures: incremental, sensor-based flight control replaces model terms with direct measurements and is robust to model

error as a consequence [9]; measurement-based (“model-less”) voltage control estimates sensitivity coefficients from data precisely because modelled ones are unreliable [10]; and online feedback-based linearised power flow updates its parameters from instantaneous measurements, already reporting robustness to measurement error, communication failure and update rate [11]. We reconfirm that principle; we do not claim it.

What has not been established is the *closed-loop consequence*. The metric in [11] is voltage estimation error, an open-loop measure of accuracy. Ours is whether the filter still refuses an unsafe action. Neither implies the other. A model can be badly wrong and still turn away everything dangerous, or mildly wrong and wave everything through. Accuracy that decays gently does not entail protection that decays gently—and we find that it does not. Our contributions are:

- 1) **A silent-failure characterisation.** A never-refreshed projection reproduces the unprojected violation rate while reporting zero infeasibilities. The failure is total, and it raises no alarm on the channel an operator would watch (Sec. V-B).
- 2) **A decomposition of the linearisation** into Jacobian and base point, each perturbed independently. The half that is expensive to obtain tolerates a factor-of-two error; the half constrained by metering carries the protection (Sec. V-C).
- 3) **A runtime diagnostic.** Two unrelated faults each produce a clean violation rate. We give the telemetry that separates a working layer from an absent one (Sec. V-D).
- 4) **Measured tolerance bands and an operating envelope**, on two feeders sharing no topology, voltage level or impedance scale.

We also report, rather than bury, that on these testbeds it is the projection and not the learned policy that delivers safe service (Sec. V-A). That finding is what turned the study towards the layer instead of the learner, and it is why a trained agent enters the experiments below as an input to the layer rather than as the object of study.

II. RELATED WORK

Safety layers for distribution voltage control. Dalal *et al.* [1] introduced a layer that analytically corrects an action to satisfy linearised constraints. The construction has since been carried into multi-agent voltage control [2], Volt-VAR optimisation [3], communication-delayed inverter control [4], physics-regularised graph learning [5] and EV charging [6]. Two of them take directly opposing positions on model sensitivity: one advertises that approximate sensitivity information suffices [2], while another motivates itself by projection being sensitive to parameter error [5]. Neither reports the experiment that would settle the disagreement. Section V-C reports it.

Sensitivity computation and currency. Analytic sensitivity coefficients for radial networks are standard [12]. Measurement-based estimation [10] and online feedback linearisation [11] tackle model unavailability head-on. Bai *et al.* [11] are the nearest prior work, and the distinction is the

one drawn above: they measure how accurate a linearisation remains, whereas we measure whether a filter built on one still refuses unsafe actions.

Delay and staleness. Communication delay has been treated as a millisecond-scale synchronisation problem to be imputed away [4]. Age of information has served as a formal staleness metric in power systems, for transmission-level load frequency control [13]. We treat hour-scale base-point age as a design variable for distribution constraint enforcement, where the failure mode is inertness rather than instability.

III. THE SAFETY LAYER AND ITS TWO INPUTS

A. Feeder and fleet

A radial feeder carries N_b buses, four of which host bidirectional charging stations rated S^{rat} . AC power flow is solved at every control step by a backward–forward sweep validated against Newton–Raphson to 10^{-9} p.u. The control step is five minutes over a 288-step day. Vehicles arrive from a bimodal profile with log-normal dwell times and per-vehicle energy targets, and a station’s set-point is shared among its connected vehicles in proportion to the energy each still owes. We follow the sign convention of the underlying solver: $p > 0$ is injection into the bus (vehicle-to-grid discharge) and $p < 0$ is charging. The operational requirement is $\underline{v} \leq v_i \leq \bar{v}$ at every bus, with $[\underline{v}, \bar{v}] = [0.95, 1.05]$ p.u.

B. The projection

Let $u = (p^{\text{req}}, q^{\text{req}})$ be the requested station set-points. The layer returns the admissible point closest to them,

$$\begin{aligned} \min_{p,q} \quad & \|p - p^{\text{req}}\|_2^2 + \|q - q^{\text{req}}\|_2^2 \\ \text{s.t.} \quad & \underline{v} + \varepsilon \leq \hat{v}(p, q) \leq \bar{v} - \varepsilon \\ & \|(p_j, q_j)\|_2 \leq S_j^{\text{rat}}, \quad |p_j - p_j^-| \leq \Delta^{\text{max}}, \end{aligned} \quad (1)$$

in which ε is a tightening margin, p^- the previous set-point, and the predicted voltage is the linearisation

$$\hat{v}(p, q) = \underbrace{v^0}_{\text{base point}} + \underbrace{\frac{\partial V}{\partial P}}_{\text{Jacobian}} (p - p^0) + \frac{\partial V}{\partial Q} (q - q^0). \quad (2)$$

Problem (1) is a second-order cone program, solved with CLARABEL through a parametrisation that is compiled once and re-solved with fresh parameters each step. A candidate is accepted on its *measured* constraint residual rather than on the solver’s status label, because a hedged-but-feasible point is common and discarding one freezes stations for nothing.

C. What ages, and what does not

Equation (2) is the object of this paper. Its two halves are not merely conceptually distinct; they are procured by different departments, on different budgets, at different cadences. The Jacobian is a property of the network. We compute it analytically from a single LU factorisation of the power-flow Jacobian, in 0.662 ms against 447.6 ms for finite differences, so a utility with a model can afford to refresh it as often as it likes. The base point (v^0, p^0, q^0) is a property of the

TABLE I
THE TWO TESTBEDS. NOTHING IS SHARED.

	MV feeder	LV feeder
Base network	IEEE 33-bus	Kerber <i>Dorfnetz</i>
Buses (incl. source)	34	117
Nominal voltage	12.66 kV	0.4 kV
Laterals	1 trunk, 3 spurs	6
Load points / total	32 / 3.72 MW	57 / 0.34 MW
Stations	4 × 80 kVA	4 × 22 kVA
Idle $V_{\min}$	0.9658 p.u.	0.9686 p.u.
Idle violations	0.0000	0.0000
Uncoord. violations	0.0626	0.0667
Uncoord. service	0.804	0.779

present operating condition, and no amount of computation will produce one. It has to be observed.

That asymmetry is why the decomposition is worth making. A designer cannot buy base-point currency with compute, which is exactly why its requirement needs to be known before the layer is relied upon. We write τ for the *base-point age*: the number of control steps since the linearisation was last taken. Throughout, “refresh interval τ ” means the Jacobian and the base point are both recomputed every τ steps and held fixed in between, which is what a controller with periodic state updates does.

D. What happens when the projection fails

When (1) admits no solution, the station holds its previous set-point, and after three consecutive failures it latches to zero. We disclose this rather than defend it: Sec. V-D shows the fall-back becoming the dominant mechanism under model error, which is a finding about the design and not an endorsement of it.

IV. EXPERIMENTAL SETUP

A. Two feeders that share nothing

Generality cannot be claimed from one set of line impedances, so every result below is measured on two networks with nothing in common (Table I). The MV testbed is the IEEE 33-bus Baran–Wu feeder [14] with a Thévenin source impedance Z_{sub} inserted upstream of the slack bus, giving 34 buses. The LV testbed is the Kerber *Dorfnetz* German village benchmark [15]: six laterals and 57 households behind a 400 kVA transformer at 0.4 kV, with line impedances an order of magnitude smaller. Both are built in PANDAPOW [16]. The LV distribution transformer is folded into an equivalent series line—exact for a fixed tap—and line shunt capacitance is neglected, which we measured to shift $V_{\min}$ by 2.2×10^{-5} p.u., three orders of magnitude below the tightening margin ε .

B. Operating points chosen by measurement

An operating point picked for convenience can trivialise a study in either direction, so both feeders are held to two requirements: the *idle* feeder must sit strictly inside the band, so that every violation is attributable to a charging decision, and an uncoordinated charger must break the band, so that

TABLE II
THE SAME PROJECTION, DIFFERENT CONTROLLERS (MV FEEDER)

Controller	Viol.	Service	Cost	ms/step
Idle (no charging)	0.0000	0.0000	\$0	1.01
Uncoordinated	0.0626	0.8041	\$686	1.04
IEEE 1547 droop	0.0000	0.0067	\$63	1.04
SafeSAC (learned + proj.)	0.0000	0.0889	\$255	—
Uncoordinated + proj.	0.0000	0.7094	\$595	3.10

the layer has something to do. On the LV feeder we swept 36 combinations of substation stiffness, background load and fleet size, then selected the cell closest to the MV feeder’s uncoordinated behaviour on *both* axes at once—violation rate and service—since a second feeder tuned harder would inflate the effect and one tuned easier would hide it.

C. Request sources

A claim about a safety layer should not depend on what feeds it. We therefore drive the layer from unrelated sources: an uncoordinated greedy charger, a deadline-aware heuristic that defers non-urgent stations, and, on the MV feeder, a soft actor–critic agent [17] trained under an augmented-Lagrangian cost constraint over 3 seeds. Three sources are used on the MV feeder and two on the LV feeder.

D. What we record

We report the *voltage violation step rate*, the fraction of control steps at which any bus lies outside the band, and *service*, the fraction of vehicles meeting their energy target at departure. Just as importantly we report the layer’s own telemetry: the fraction of steps at which (1) was infeasible, and the fraction at which a station sat latched to zero. Section V-D is about why the first two do not suffice.

Each cell is 25 evaluation episodes on a shared seed band, so that methods are paired episode by episode. Seeds are derived by hashing run labels, with disjoint training and evaluation ranges. The implementation carries 90 tests and reproduces the reference results of the underlying testbed exactly.

V. RESULTS

A. The layer, not the learner

Before asking what the layer requires, it is worth establishing that the layer, rather than the policy above it, is what delivers safe service. Projection is almost always composed with a learned policy, which quietly conflates two questions: whether the layer makes actions safe, and whether learning adds anything on top of it. Applying the identical projection to a zero-intelligence greedy charger separates them (Table II).

The projection applied to a controller with no learning at all delivers $8.0\times$ the service of the same projection applied to the trained agent, at identical zero violations. A sweep over background load and EV penetration found no cell in which deferring non-urgent stations beat charging everything: on this testbed deferred energy is never recovered, so the binding resource is energy over the day rather than its allocation across

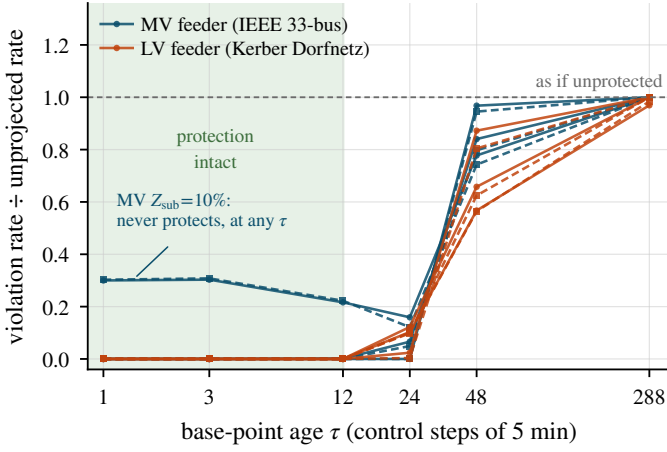


Fig. 2. Violation rate as a fraction of each cell’s own unprojected rate, for twelve cells across two feeders, three substation stiffnesses each and two request sources. Shading marks the ages at which every cell inside the operating envelope still holds the band to within 10^{-3} . The two MV curves that never reach zero are at a stiffness where projection alone cannot hold the band at any cadence.

stations, and there is no scheduling structure left for a policy to exploit. *This is a statement about this testbed at this training budget, not a claim that learning cannot help.* It is why the rest of the paper interrogates the layer, and why the trained agent appears below only as one request source among three.

B. Ageing the base point

We now hold the network model exactly correct and vary nothing but τ . Fig. 2 shows every cell normalised by its own unprojected rate, so that two feeders whose absolute rates differ threefold can be read on one axis; Table III gives the same data in absolute terms, for the uncoordinated source. Three things stand out.

The layer switches off rather than degrading. At $\tau = 288$ —one refresh per day, which is to say never after the first solve—the violation rate is the *unprojected* rate, recovering 96.9–100.0% of it across all 12 cells and both heuristic sources. The trained agent behaves the same way: 97.9–99.6% recovery over 6 seed-and-stiffness cells. A base point captured on an idle feeder makes the constraint look slack at every subsequent step, the feasibility pre-check skips every solve, and each request passes through untouched.

TABLE III
VIOLATION STEP RATE AGAINST BASE-POINT AGE, UNCOORDINATED SOURCE. CORRECT NETWORK MODEL THROUGHOUT. BOLD MARKS EXACTLY ZERO.

Feeder	Z_{sub}	No	Base-point refresh interval (control steps)					
		proj.	1	3	12	24	48	288
MV	6%	0.0606	0.0000	0.0000	0.0000	0.0000	0.0586	0.0606
	8%	0.1051	0.0000	0.0000	0.0001	0.0069	0.0883	0.1051
	10%	0.1361	0.0408	0.0413	0.0294	0.0217	0.1058	0.1361
LV	1%	0.0642	0.0000	0.0000	0.0000	0.0015	0.0560	0.0642
	2%	0.1206	0.0000	0.0000	0.0000	0.0147	0.0793	0.1206
	3%	0.2076	0.0000	0.0000	0.0000	0.0214	0.1178	0.2013

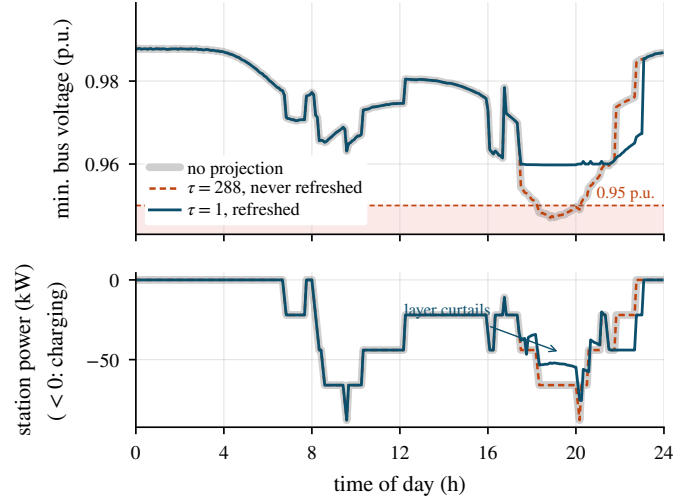


Fig. 3. One episode on the LV feeder, three controllers. The never-refreshed projection (dashed) lies inside the unprojected trace (grey) everywhere, in voltage and in dispatch alike, while the refreshed projection curtails through the evening peak and holds the band.

Fig. 3 shows this happening within a single episode, which aggregated rates cannot. The never-refreshed run and the entirely unprojected run are not merely similar; over the whole day their minimum-voltage trajectories differ by at most 1.4×10^{-6} p.u., and both spend the same 24 of 288 steps below the limit. The layer is not intermittently failing. It is absent.

It is silent while doing so. The infeasibility rate at $\tau = 288$ is 0.0000 in every cell. The layer is not failing to solve; it never solves. An operator watching the component’s own fault channel sees nothing at all.

The transition is a cliff, and its location is not a constant. Between one interval and the next, protection goes from complete to essentially absent. On the MV feeder the last age at which every cell holds the band exactly is 24 steps at $Z_{\text{sub}} = 6\%$, falling to 3–12 steps at 8%; on the LV feeder it is 12 steps at every stiffness tested. We therefore decline to offer a universal refresh figure. *The cliff exists everywhere we looked; where it sits varies with the deployment and has to be measured for one.* Nor do we claim to know what sets it: on the MV feeder it tracks substation stiffness, and on the LV feeder it does not move at all.

C. Corrupting the model

We now invert the experiment: the base point is measured on the real feeder at every step, and only the Jacobian is corrupted, by computing it on a feeder whose line impedances have been scaled by a factor k . This is the arrangement a deployed controller with an out-of-date network model but live telemetry actually occupies. Scaling impedances rather than substituting a different feeder keeps the Jacobian’s dimensions matched while moving its magnitude across $0.20 \times -5.67 \times$ (Fig. 4).

A Jacobian wrong by up to roughly a factor of two costs nothing: violations are zero throughout $0.80 \times -2.06 \times$ on both feeders and both heuristic sources. Set against Sec. V-B, the

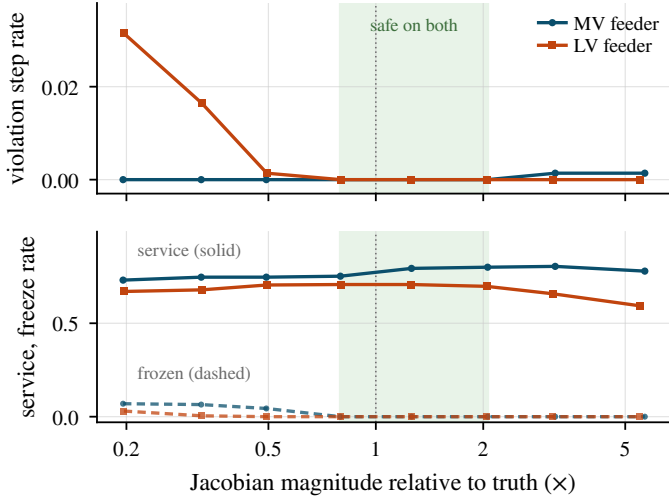


Fig. 4. Violation rate, service and freeze rate against Jacobian magnitude, with the base point measured every step. Shading marks the window in which violations are zero on both feeders and both heuristic sources. Under-estimated impedance costs service through the fallback long before it costs voltage.

TABLE IV
TWO FAILURE MODES. NEITHER SHOWS IN THE COLUMN AN OPERATOR WATCHES.

Condition	Viol.	Infeas.	Frozen	Serv.
MV No projection	0.0560	—	—	0.8642
MV Correct model, $\tau=1$	0.0000	0.0016	0.0000	0.7698
MV Jacobian $0.2\times$ truth	0.0000	0.0833	0.0694	0.7309
LV No projection	0.0642	—	—	0.7535
LV Correct model, $\tau=12$	0.0000	0.0000	0.0000	0.7304
LV Never refreshed, $\tau=288$	0.0642	0.0000	0.0000	0.7535

asymmetry is the paper’s central practical point. The half of the linearisation that is expensive to obtain, and that a distribution utility may simply not possess, tolerates substantial error. The half that cannot be computed at all carries the protection.

Outside that window the failure is directional, and which direction is feeder-dependent: the LV feeder leaks violations when impedance is under-estimated, the MV feeder when it is over-estimated. Over-estimation is the conservative side, and it is not free either—on the LV feeder a $5.5\times$ Jacobian holds violations at zero but costs 11 percentage points of service.

D. Zero is not evidence

The two failure modes are mirror images, and Table IV sets them side by side.

A stale base point yields a clean *infeasibility* rate while the feeder goes entirely unprotected; on the LV feeder the never-refreshed row and the no-projection row are the same number, 0.0642, to four decimal places. An under-estimated impedance yields a clean *violation* rate—on the MV feeder exactly zero, apparently perfect—while the projection is infeasible on 8.3 % of steps and the fallback has latched stations to zero on 6.9 %, at a cost of 3.9 percentage points of service. In that second case the layer is not protecting the feeder. It is switching the load off, and being credited for the result.

Neither condition is detectable from the outcome alone. We therefore suggest that a deployed projection expose, and an operator alarm on, three quantities rather than one:

- 1) **Base-point age** τ , alarmed against a threshold measured for that feeder rather than assumed for it.
- 2) **Solve rate**. A projection reporting zero infeasibilities and a near-zero solve rate on a loaded feeder has stopped binding. It is not succeeding; it is absent.
- 3) **Fallback activation**. A non-trivial latch rate means the reported violation rate was paid for by curtailment rather than by projection.

The first two together amount to a one-line assertion that any of the works in Sec. II could adopt at no cost.

E. What it costs to run

The analytic sensitivity computation takes 0.662 ms against 447.6 ms for finite differences, and a full projection step costs a few milliseconds against a 300 s control interval—a duty cycle near 0.004 %. Every refresh interval examined here is computationally free. The constraint on base-point currency is metering, not computation, which is the practical reading of Sec. V-C.

VI. DISCUSSION AND LIMITATIONS

Scope. These results are for weak, low-short-circuit-ratio radial feeders, where voltage rather than thermal loading is the binding constraint. On robustly built distribution networks the reported voltage impact of EV charging is small and transformer thermal limits bind first. We make no claim there, and our testbeds are chosen accordingly.

What “refresh” means here. We simulate base-point age by recomputing the power flow every τ steps, not by modelling a metering and communication pipeline. The axis is the *age of the base point*, agnostic to whatever produces it. Whether real deployments occupy the unsafe region is a question this paper does not answer; we observe only that the LV feeder’s cliff at one hour coincides with common advanced-metering reporting intervals.

Statistics. We report measured rates over 25 evaluation episodes per cell and claim no confidence intervals. The effects reported here are of the form 0.0000 against 0.0642. The sub-percentage-point differences we observed between short refresh intervals were of opposite sign on the two feeders, and we therefore make no claim about them.

The fallback. Freeze-to-zero is a poor fallback, as Sec. V-D makes plain. A layer that curtails to satisfy a constraint it cannot verify converts a voltage problem into a service problem without saying so. Designing a fallback that degrades in a declared direction is open work.

Future work. The natural next step is to make τ an explicit argument of the constraint, tightening the band by a data-driven bound on net-load drift over τ so that protection degrades continuously instead of switching off. The construction exists in the control literature for systems with sporadic measurements [18]; adapting it to an AC power-flow band,

with a drift bound estimated from measured load rather than assumed adversarial, is ongoing.

VII. CONCLUSION

A sensitivity-based safety layer's protection rests on the currency of its linearisation base point rather than on the accuracy of its network model. Wrong by a factor of two, the model costs nothing. A base point that stops being refreshed removes the protection entirely, reproducing the unprojected violation rate while reporting no fault, on two feeders that share neither topology nor voltage level nor impedance scale. The transition is a cliff whose location varies between deployments, and two distinct faults both present as a clean number in the column an operator would watch. The remedy is not a faster refresh rate—every interval examined here is computationally free—but instrumentation: measure the base-point age, the solve rate and the fallback activation, and alarm on all three.

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